

Hengnan Ma

929-329-4131 | hengnanma@gmail.com | [Linkedin](#) | [Github](#)

EDUCATION

The Cooper Union for the Advancement of Science and Art

Manhattan, NY

Bachelors of Science in Engineering, Minor in Computer Science

EXPERIENCE

Konsilix LLC |

May 2024 – Present

Software Engineer | Java, SpringBoot, Kubernetes, Github Actions, AWS

NY

- Developed and maintained distributed, **microservice** based software solutions built with Apache Zookeeper, implementing test-driven development methodologies to ensure code reliability and maintainability.
- Strengthened system security by integrating authentication mechanisms using **OAuth2 and JWT tokens**, enabling users access via GitHub, Google, and other third-party providers.
- Participated in code reviews, debugging, and functional testing to enhance software quality.
- Leveraged **Kubernetes** with **Helm Charts** for efficient and reusable deployments, resulting in a increase in team productivity.
- Implemented automation scripts and scalable **CI/CD pipelines** using **GitHub Actions**, automating building, testing, and deployment, reducing release cycles and deployment errors.
- Followed **Agile** development methodologies to collaborate effectively with cross-functional teams.

PROJECTS

Rating Platform | Java, SpringBoot, React, PostgreSQL, Docker, Firebase, AWS

Github

- Designed and launched a full-stack web application from scratch using **Spring Boot, React.js, PostgreSQL, Firebase, and Figma**.
- Leveraged Feedback from 30+ active users to build a scalable backend architecture with REST APIs and Amazon RDS, ensuring seamless cross-platform functionally, data synchronization, and increasing user satisfaction by 80%.
- Developed and tested **15+ RESTful APIs** and rigorously tested with **Postman**.
- Containerized with **Docker** and deployed on **AWS EC2** and **S3**, leveraging **auto-scaling groups, load-balancing, and disaster recovery**, achieving 99.9% uptime.

Taxi-Fare Predictor with Machine Learning | Pytorch, AWS Services, Bash

Github

- Built a **machine learning ETL pipeline** to predict taxi fares for ride-hailing services, processing and validating a dataset of **53.2 million** rows with **AWS Glue and Athena**, reducing data query times by 98% through **parquet** transformation.
- Designed and implemented a **distributed training pipeline** on a cluster of **10 EC2 nodes**, accelerating training time by 800% compared to single-node setups.
- Achieved a mean absolute error of \$1 with a regression model in **PyTorch**, improving prediction accuracy by 300% over baseline models.

Parking Spot Finder | Python, Flask, React, MongoDB, Git

Github

- Designed and developed a **Python** based **RESTful API** backend using **Flask** and **SQLAlchemy**, incorporating **React.js** to provide an intuitive user interface.
- Implemented user registration, login, and profile management with automated SQL queries, including rule-based access for admin users.

TECHNICAL SKILLS

Frontend: React, CSS, HTML, JavaScript

Backend: Node.js, Python, Java, C++

Cloud & DevOps: AWS, Azure, Kubernetes, GitHub Actions

Database: MongoDB, SQL, NoSQL, Postgresql, DynamoDB

Frameworks & Tools: PyTorch, Spring Framework, Pandas, Postman, Bash, PowerShell

CERTIFICATIONS

Amazon Web Services DevOps Engineer - Professional

Amazon Web Services Certified Developer – Associate